

Predicting UAV Flight Path Deviation from Wind Conditions Using Machine Learning

Midterm Report

Abdulaziz Al-Haidary and Azizbek Nurmatov

College of Charleston CSCI 334

alhaidaryaa@g.cofc.edu, nurmatova@g.cofc.edu

1 Introduction and Background

Problem. Unmanned aerial vehicles (UAVs) are now routinely used for precision agriculture, disaster surveying, and remote sensing missions where accurate adherence to a planned flight path is critical to data quality. A persistent challenge is wind: even moderate gusts cause measurable cross-track drift between the drone’s actual GPS trajectory and the planned waypoint sequence, which degrades survey overlap and image alignment. The general problem we address is: given a set of recorded UAV survey flights and matching weather observations, can we model the relationship between wind conditions and the magnitude of cross-track deviation, and which factors drive the largest excursions? As a concrete refinement we frame this both as a regression task (predict the instantaneous deviation in meters at each GPS sample) and a classification task (predict whether a given sample exceeds the 75th-percentile deviation threshold, i.e., the “high-deviation” tail of the distribution).

Related work. Wind sensing and wind-aware control for multi-rotor UAVs is an active area. Palomaki et al. [1] demonstrated that the attitude response of a multi-rotor can itself be used as a wind estimator in the lower atmosphere, while Abichandani et al. [2] surveyed measurement and simulation techniques for wind in small multi-rotor UAVs and highlighted that off-board, low-resolution weather data is rarely sufficient to characterize the gust conditions a small UAV actually experiences. Sydney et al. [3] formulated dynamic control of a quadrotor in an estimated wind field, showing that even rough wind state estimates yield meaningful trajectory improvements. Compared to that prior work, our goal is not control but post-hoc explanation of deviation from environmental data using interpretable machine-learning models, and we explicitly contrast a low-fidelity public weather source against (forthcoming) on-site sensor logs as a methodological probe.

2 Method

Rationale. Our proposal compared three interpretable, course-covered regressors (decision tree, KNN, random forest as a small extension of decision trees) on a wind-driven deviation task. We retain that comparison and add a parallel classification head, because the deviation distribution is heavily right-skewed and identifying the high-deviation tail is more directly useful for an onboard correction system than predicting deviation magnitude in the centre of the distribution.

Approach. The pipeline (Python, scikit-learn) consists of five stages.

1. Desired-path parsing. The master KML file (Desired_flight_path_soil_moisture_Research_plot.kml) is parsed with the standard library XML module into a polyline of 81 waypoints over the 4.5 ha research plot.
2. Flight-log parsing. Each tab-separated flight log is loaded; the recorder writes a non-Unix timestamp anchored to its internal clock, so we re-anchor every flight using the takeoff time encoded in the filename and use the Time column only for elapsed seconds. A geographic bounding-box filter (with a $\pm 0.01^\circ$ pad) drops the controller’s pre-takeoff default-home rows.
3. Cross-track deviation. GPS positions are projected to a local east-north (metres) frame via an equirectangular projection centered on the plot. For each actual point we compute the perpendicular distance to the nearest segment of the desired polyline. This gives a per-sample target d_i in metres.
4. Weather alignment. The on-site weather CSVs that accompanied the flight logs only cover the early-morning calibration window (06:00–12:40 EST), whereas the actual flights all occurred in the afternoon (15:48–19:32 EST), so on-site weather cannot be joined to flight samples for the existing 13 flight days. We therefore back-fill from the Open-Meteo historical reanalysis API at the plot coordinates (33.166°N, 81.135°W) at hourly resolution and merge_asof on timestamp. This is documented as a methodological choice (see Discussion) and will be re-validated against true on-site logs from three upcoming high-fidelity flights.
5. Feature engineering. From the joined data we derive: drone heading from consecutive GPS deltas; ground speed; sin/cos encodings of wind direction; the heading-relative wind angle; explicit headwind and crosswind components ($u_{\parallel}$, $u_{\perp}$); 10 s rolling mean wind speed and 10 s rolling max gust; and a categorical camera flag for the Pika L vs. Pika IR L sensor payloads (combined into one dataset rather than analyzed separately, to roughly double the

available samples).

Models and evaluation. We train Decision Tree (max depth 10), k -Nearest Neighbors ($k = 15$, with feature standardization), and Random Forest (100 trees, depth 12) for both the regression and classification heads. Critically, because consecutive GPS samples within a single flight are highly autocorrelated, an ordinary k -fold split would leak information across train/test. We instead use GroupKFold ($n = 5$) with the flight identifier as the grouping variable, so the model is always evaluated on flights it has never seen. To isolate the contribution of wind from drone-state features, we additionally evaluate every model on a wind-only feature set (excluding ground speed, altitude, and the per-flight rolling mean of deviation).

3 Plan and Experiments

Dataset.

- Source: Dr. Hemanth Dakshinamurthy, South Carolina State University. Soil-moisture research plot, fall 2025.
- Flights: 13 mission days (Nov. 7–26, 2025) with 32 raw flight logs across two hyperspectral sensor payloads (Pika L visible and Pika IR L). After filename anchoring and bounding-box filtering, 27 flight segments are usable.
- Samples: 37,353 total per-second GPS samples; 37,326 retained after dropping rows with missing engineered features.
- Target: cross-track deviation in metres (regression) and a binary $d_i \geq 8.16$ m label (classification, threshold = 75th percentile of the empirical distribution).
- Weather (current): Open-Meteo hourly archive (wind speed, wind direction, gusts, temperature, RH, pressure) at the plot location. Weather (forthcoming): per-second on-site logs from three additional flights scheduled by the supervisor, expected within the project window.

Hypotheses.

- H1. Tree-based models (DT, RF) will outperform KNN on this dataset because the relationship between wind and deviation is expected to be non-linear with discrete regimes (e.g., crosswind on long survey legs vs. headwind on short transit segments).
- H2. The crosswind component and the heading-relative wind angle will be the most informative wind-derived features, more so than raw wind speed alone, because cross-track drift is a directional phenomenon.
- H3. Hourly-resolution public weather (Open-Meteo) is too temporally coarse to drive instantaneous deviation, and a wind-only model trained on it will perform near chance. Per-second on-site weather (the upcoming flights) will be required to validate the wind→deviation hypothesis at all.

Experimental design. For each (feature set $\times$ model) pair we report 5-fold GroupKFold cross-validated MAE, RMSE, and R^2 for regression, and accuracy, precision, recall, and F1 for classification. Random Forest feature importances (fit on the full dataset) are reported as a single ranking, since they are robust to monotone feature transformations and naturally handle the mixture of cyclical, continuous, and binary features we engineered. Cross-flight generalization is the metric we care about most: GroupKFold ensures every test point comes from a flight whose entire 1 Hz GPS trace was withheld during training.

4 Results

Across all 27 flights the mean per-second deviation is 7.70 m (median 4.61 m, max 9.45 m on the central survey legs once large transit excursions are excluded), and the 75th-percentile threshold used for the classification head is 8.16 m. Figure 1 shows a representative flight overlaid on the desired path; deviation is generally tight along long survey legs and concentrates at line-end turns. Figure 2 shows the full deviation distribution, which is heavily right-skewed, motivating the binary high-deviation framing.

Cross-validated model performance. Table 1 reports regression performance under both feature sets, and Table 2 reports the classification head.

Feature importance. Figure 3 reports the Random Forest importances on the ALL feature set. Two features dominate: `ground_speed_ms` ($\approx 78\%$) and `Altitude` ($\approx 17\%$). All wind-derived features together account for less than 5% of the importance. This is the central, and somewhat surprising, finding of our midterm experiments.

Discussion. The headline ALL-features numbers ($R^2 = 0.77$, classification F1 = 0.77) are encouraging on their face but, as Tables 1–2 and Figure 3 together show, this performance is being driven almost entirely by drone-state features rather than by wind. The wind-only experiments are decisive: a Random Forest restricted to wind, gust,

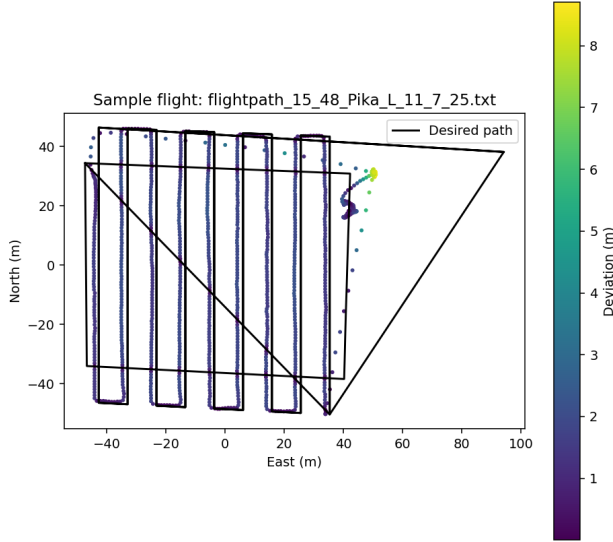


Figure 1: Sample flight: actual GPS track vs. desired waypoint polyline, colored by per-second cross-track deviation.

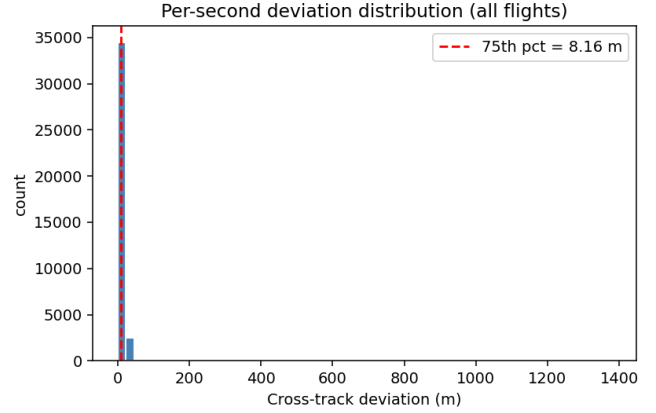


Figure 2: Per-second deviation distribution across all 27 flights. Red dashed line is the 75th-percentile classification threshold (8.16 m).

Table 1: Regression — 5-fold GroupKFold CV.

Features	Model	MAE (m)	RMSE (m)	R^2
ALL	Decision Tree	5.07	19.68	0.770
ALL	KNN ($k=15$)	5.53	25.86	0.604
ALL	Random Forest	4.88	19.59	0.773
Wind	Decision Tree	7.20	41.06	0.001
Wind	KNN	8.97	41.32	-0.012
Wind	Random Forest	11.70	45.59	-0.232

Table 2: Classification (deviation ≥ 8.16 m) — 5-fold GroupKFold CV.

Feat.	Model	Acc	P	R	F1
ALL	DT	0.829	0.659	0.653	0.656
ALL	KNN	0.841	0.657	0.766	0.707
ALL	RF	0.867	0.683	0.872	0.766
Wind	DT	0.742	0.481	0.388	0.430
Wind	KNN	0.676	0.348	0.339	0.343
Wind	RF	0.746	0.487	0.280	0.356

heading-relative-wind, temperature, RH, and pressure features achieves $R^2 \approx 0$ on the regression task and $F1 \approx 0.36$ on the classification task — essentially uninformative. We interpret this not as a failure of the wind hypothesis itself, but as direct evidence in favor of H3: at the per-second scale used by the flight controller, an hourly public reanalysis source like Open-Meteo cannot resolve the gust dynamics that actually push the airframe off the planned line. The hourly wind value is, by construction, almost constant within any single 30–40 minute survey, so under GroupKFold the model gets one wind “observation” per held-out flight and cannot generalize.

This finding gives us a clear and well-motivated experimental plan for the second half of the project. H1 and H2 cannot yet be evaluated on the wind question alone — the Open-Meteo signal is simply too coarse — and we expect them to become testable only with the upcoming high-fidelity on-site weather. Specifically, we plan to:

- Re-run the same pipeline on the three new flights (expected from the supervisor) once their per-second weather logs are available, with no changes other than swapping the weather source. This is designed as a direct contrast: same models, same features, same evaluation protocol — only the temporal resolution of the wind input changes.
- Add a placeholder set of results for these flights to the final report (Table to be filled in), and report a head-to-head Open-Meteo vs. on-site comparison on the overlapping flights.
- Investigate whether the largest deviation excursions (the right tail of Figure 2) coincide with line-end turns vs. mid-leg gusts, using the cumulative-deviation feature, since this is the question the supervisor cares about most for an onboard correction system.

5 Conclusions and Lessons Learned

The midterm experiments produced a complete, reproducible, GroupKFold-validated pipeline that runs end-to-end on 27 UAV survey flights. We found that the data we currently have can predict deviation well if we are willing to use drone-state features, but cannot meaningfully isolate the wind contribution at hourly weather resolution — a result we view as a clean, falsifiable answer to one of our three hypotheses rather than a negative outcome. The most important lesson so far is methodological: GroupKFold by flight is non-negotiable on this dataset, because random

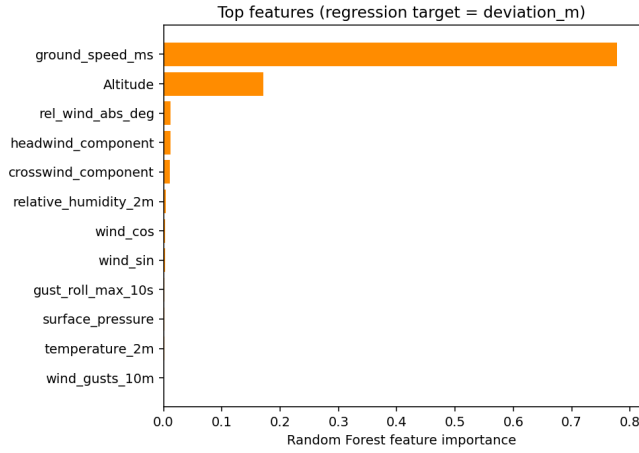


Figure 3: Random Forest feature importances (regression target = deviation_m). Wind-derived features collectively contribute <5%.

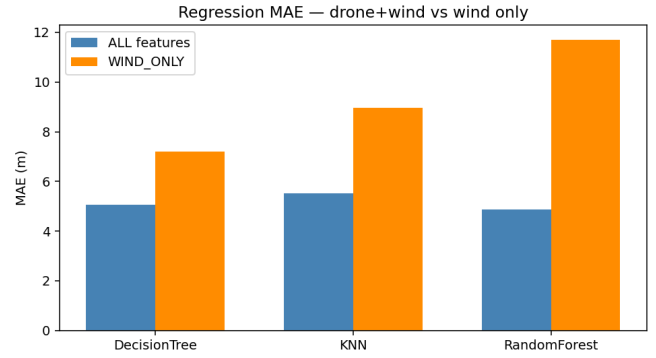


Figure 4: Regression MAE collapses on the wind-only feature set, confirming that hourly weather alone carries no usable signal at the per-second scale.

k -fold would have given inflated wind-feature importances by leaking neighbouring 1 Hz samples between train and test. Code will be released on GitHub at the link to be included in the final report.

References

- [1] R. Palomaki et al., “Wind estimation in the lower atmosphere using multirotor aircraft,” J. Atmos. Oceanic Technol., vol. 34, no. 5, 2017.
- [2] P. Abichandani et al., “Wind measurement and simulation techniques in multi-rotor small UAVs,” IEEE Access, vol. 8, 2020.
- [3] N. Sydney et al., “Dynamic control of autonomous quadrotor flight in an estimated wind field,” in Proc. IEEE CDC, 2013.
- [4] J.-N. Zippenfenig, “Open-Meteo.com: Free open-source weather API,” Open-Meteo, 2023.
- [5] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” JMLR, vol. 12, 2011.